

When the vessel breaks

What to say, slide by slide • for an academic physiotherapy audience

Sunday 4 October 2026 • 8 pm

⌚ About 21 minutes, then questions

36 slides + 2 hidden

THE BIG IDEA

ICH is a process, not a moment.

- 1** The vessel decides where it bleeds.
- 2** The first hours decide how big it gets.
- 3** The blood keeps injuring the brain for days.

KNOW THESE BY HEART

Three lines to own

FIRST LINE	"My chapter is Chapter 5... I want to start with one person."
HEADLINE	"ICH is a process, not a moment."
LAST LINE	"Thank you. Which claim tonight would you challenge?"

HOW TO READ EACH SLIDE

Key

KEY POINT the one idea the room must leave with

SAY talking points, with the words to land in **bold**

WORD FOR WORD learn these exactly

⏏ Pause 🖐 Ask the room 📍 Point

⦿ Prop (the golf ball) ⤵ Slow down

» the line that carries you to the next slide

🗉 **PLAIN WORDS** the same point for junior colleagues

🔍 **IF CHALLENGED** appraisal point; not in the timed script

Running order

	SECTION	SLIDES	TIME
00	Opening Story first, then the physiotherapy lens and four objectives.	Slides 1–6	3:35
01	The vessel Why a small artery gives way, and what its location predicts.	Slides 7–15	5:30
02	The first hours Rupture, growth and pressure. The golf-ball moment lives here.	Slides 16–21	3:25
03	The days after How the blood itself keeps injuring the brain.	Slides 22–26	2:50
04	At the bedside When to start, a Day-4 case, the recovery debate, then close on Jill.	Slides 27–34	5:50
	Questions Leave the references up. Hidden backups on slides 37 and 38.	Slides 35–36	Rest of slot

Four moments with the room

- **Slide 14, ~8 min:** 30 seconds with a neighbour: "Primary or secondary?"
- **Slide 18, ~10 min:** hands up for A, B or C before the dots reveal.
- **Slide 25, ~14 min:** hands up: "true or false?"
- **Slide 29, ~16½ min:** the Day-4 case: A, B or C.
- Always **say what you see**. Online: use the chat.

The golf-ball moment

- Ball in a pocket you can reach without looking.
- On slide 20, take it out **before** you speak.
- Hold it at shoulder height. Three seconds of silence.
- Then: "Jill's surgeons removed a clot this size."
- Put it down where everyone can still see it.

Before you start

- Presenter View on; clicker tested; water within reach.
- Slides saved in three places; PDF on your phone.
- Your three messages on a card, in case the tech fails.
- Talk to two or three people before you begin.
- Feet planted, one slow breath out, find a friendly face.

If things go wrong

- **Lost your place:** read the slide title aloud and carry on.
- **Running long:** hide slides 10, 13, 17 and 24 (about 19 min). Shorter still: also hide 25–26 (about 17½ min).
- **Tech fails:** give the three messages and Jill's story from your card.
- **Hostile challenge:** thank them, name the design and its limit, give your position.
- **Don't know:** "I don't want to guess. I'll check and come back to you."



Slide 1

⌚ 0:20

01 When the vessel breaks

🔊 WORD FOR WORD

KEY POINT

Introduce the chapter and the physiotherapy lens, then turn to one person.

SAY

- Good evening. My chapter is **Chapter 5** of the Oxford Textbook of Stroke: the pathophysiology of non-traumatic intracerebral haemorrhage.
- Tonight I want to read it **as physiotherapists**: what the mechanism tells us about the patient in front of us.
- But I want to start with **one person**.

🔊 **PAUSE** Walk to the centre and stand still before you say a word.

» Advance to the date.



Slide 2

⌚ 0:25

02 10 December 1996

KEY POINT

A brain scientist wakes up with a bleed of her own.

SAY

- **10 December 1996. Seven in the morning.**
- A 37-year-old scientist wakes up with a **pounding pain behind her left eye**. She thinks it's a headache and climbs onto her exercise machine anyway.
- Her name is **Jill Bolte Taylor**. She is a neuroanatomist. She studies the brain for a living.

🔊 **SLOW DOWN** Tell it like a story. Slow down.

🔊 **PAUSE** Pause after "for a living".

» Advance: the red dot grows.



Slide 3

🕒 0:40

03 Within four hours

KEY POINT

In four hours she lost her balance, her arm and her speech.

SAY

- Over the next **four hours**, she watches her own brain fail. First her balance, then her right arm, then her speech.
- She tries to call for help, and the numbers on the phone look like **squiggles**.
- A blood vessel had **burst** in the left side of her brain.

👉 **ASK THE ROOM** Show of hands: "Who here has treated someone in the first week after a bleed like this?" Say what you see.

» *"Tonight is about what happened inside Jill's head, hour by hour."*



Slide 4

🕒 0:35

04 40%

KEY POINT

Four in ten die within a month, and that did not improve for decades.

SAY

- **Forty percent.** That is how many people die within a month of an intracerebral haemorrhage.
- That number **did not move between 1980 and 2008**, while treatment for ischaemic stroke was transformed.
- Part of the problem: we treat a bleed as **one moment**. The vessel bursts, the damage is done, and we wait.

👉 **POINT** Point at the number and stay silent for two seconds.

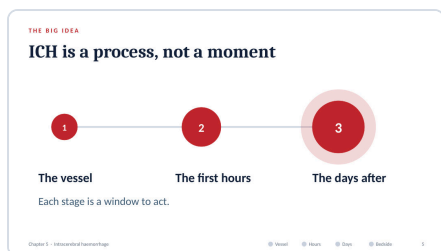
» *"Chapter 5 gives us a different view."*

🗣️ PLAIN WORDS

Four in ten people with a brain bleed die within a month.

❓ IF CHALLENGED

van Asch pooled 36 population studies up to 2008. Later registries show modest gains. Part of the old figure may be a self-fulfilling prophecy from early withdrawal of care (Becker 2001).



Slide 5

⌚ 0:40

05 ICH is a process, not a moment

🔊 WORD FOR WORD

KEY POINT

The headline of the whole talk.

SAY

- **ICH is a process, not a moment.**
- The **vessel** decides where it bleeds. The **first hours** decide how big it gets. Over the following **days**, the blood itself keeps injuring the brain.
- Each stage is a **window where we can act.**

🔊 **PAUSE** Say the headline slowly, then pause for two seconds.

» "Here's what you'll be able to do by the end."

TONIGHT
Four things you will be able to do

Explain	why a bleed keeps injuring the brain
Predict	the impairment and the risk of decline
Appraise	the strength of each claim
Apply	time and pace therapy to the stage

Chapter 5 - Intracerebral hemorrhage

⏮ Vessel ⏪ Hours ⏩ Days ⏭ End

Two ways to listen

THE TITLE
Is the message: New to stroke?
Hold on to it.

THE TAG
Is the evidence: Experienced?
Challenge it.

Slide 6

⌚ 0:55

06 Four things you will be able to do

KEY POINT

Set four objectives, and tell the room how to listen.

SAY

- By the end you will be able to **explain** why a bleed keeps injuring the brain, **predict** the impairment and the risk of decline, **appraise** the strength of each claim, mine included, and **apply** it by timing therapy to the stage.
- There are two ways to listen. **The title is the message.** New to stroke? Hold on to the titles.
- **The grey tag is the evidence:** the study design and its main limit. Experienced? Challenge the tags. I'll take challenges at the end.

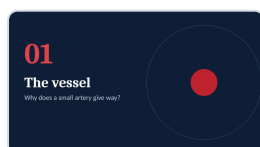
🔊 **POINT** Point at the dark card, then at a real tag on the next content slide.

» "Stage one."

01 The vessel

Why a small artery gives way, and what its location predicts.

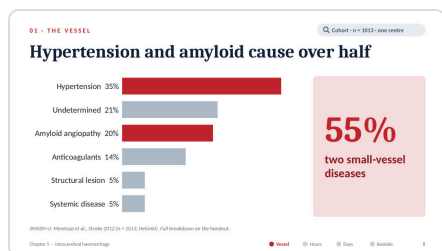
⌚ 5:30



Slide 7

⌚ 0:05

- "Stage one. **The vessel.** Why does a small artery give way?"



Slide 8

0:40

08 Hypertension and amyloid cause over half

KEY POINT

Two small-vessel diseases cause more than half of all bleeds.

SAY

- This is **SMASH-U**: just over a thousand patients in Helsinki, each sorted by cause.
- Hypertension **35 percent**, amyloid angiopathy **20**. Together more than half, and both are diseases of small vessels. **One in five** had no clear cause.
- Why should we care? The **cause predicts where it bleeds**, and where it bleeds predicts what we'll see.

➤ **POINT** Point at the two red bars and stay silent for two seconds.

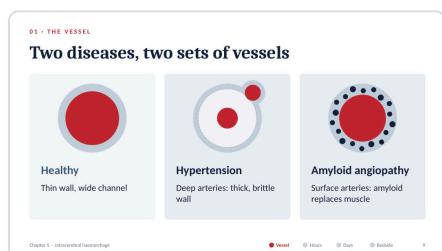
» "So what do these two diseases do to a small artery?"

PLAIN WORDS

Two diseases of tiny arteries cause most bleeds: high blood pressure and amyloid.

IF CHALLENGED

One centre in Helsinki.
"Undetermined" shrinks the harder you look, so the split depends on the work-up.



Slide 9

0:55

09 Two diseases, two sets of vessels

KEY POINT

Hypertension damages deep arteries; amyloid damages surface arteries.

SAY

- Left: a **healthy** artery. Thin wall, wide channel.
- Middle: **hypertension** attacks the tiny perforating arteries deep in the brain. The wall turns thick and glassy, but **brittle**. The bulge is a Charcot-Bouchard microaneurysm.
- Right: **amyloid angiopathy**. Beta-amyloid replaces the muscle in arteries on the surface of the brain. Mostly over 55.

➤ **POINT** Walk them left to right: say what it is, point at what matters, say what it means.

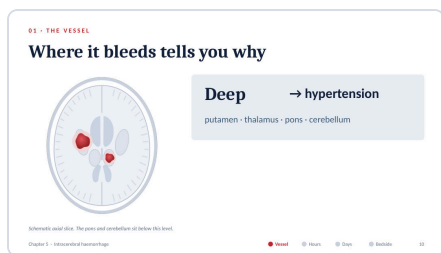
» "That gives us a rule you can use tomorrow."

PLAIN WORDS

High blood pressure makes deep arteries stiff and brittle. Amyloid makes surface arteries weak.

IF CHALLENGED

Charcot-Bouchard microaneurysms were described in 1868, but some later proved to be artefacts, so their role in rupture is still debated.



Slide 10

🕒 0:25

10 Where it bleeds: deep

KEY POINT

Deep bleed: think hypertension.

SAY

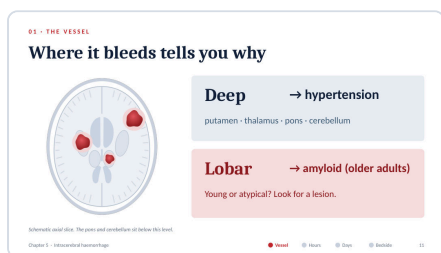
- **Where it bleeds tells you why.**
- These two are **deep**: the putamen and the thalamus. The pons and cerebellum count too.
- These arteries branch at **right angles** off big, high-pressure trunks, so they take the full force of the pressure.

➤ **POINT** Point to the two deep bleeds.

» *Advance: a third bleed appears near the surface.*

PLAIN WORDS

A deep bleed usually means a blood-pressure disease.



Slide 11

🕒 0:25

11 Where it bleeds: lobar

KEY POINT

Lobar bleed in an older adult: think amyloid. Young or atypical: be suspicious.

SAY

- This one is **lobar**: the cortex and the white matter just beneath it.
- In an **older adult**, think amyloid. In a **younger** person, or when something looks unusual, be **suspicious**.

» *"Keep that thought. But first: what does location mean for us?"*

PLAIN WORDS

A bleed near the surface in an older person usually means amyloid. In a young person, look for a hidden cause.

01 - THE VESSEL - WHAT YOU WILL SEE

Location predicts the impairment

Putamen, internal capsule	Opposite-side weakness. Clot on the PLIC: worse motor outcome.
Thalamus	Opposite-side sensory loss, ataxia; later, central pain
Cerebellum	Same-side ataxia, truncal sway. Drowsy = emergency.
Pons	Often severe: four-limb weakness, cranial nerve signs
Lobar	By lobe: weakness, neglect, aphasia, field loss; seizures

Chapter 5 - Intracerebral haemorrhage

Slide 12

🕒 1:00

12 Location predicts the impairment

KEY POINT

Where the bleed sits predicts what you will see on assessment.

SAY

- **Putamen and internal capsule:** weakness on the opposite side. Check the **PLIC**, where the motor fibres run. Clot on it: worse motor outcome.
- **Thalamus:** opposite-side sensory loss and ataxia; weeks later, sometimes central post-stroke pain.
- **Cerebellum:** same-side ataxia and truncal sway. It can block CSF and press on the brainstem, so a **drowsy cerebellar patient is an emergency**.
- **Pons:** often severe: four-limb weakness, cranial nerve signs. **Lobar:** by lobe, plus more seizures because the cortex is involved.

🔴 **SLOW DOWN** This is the slide physiotherapists will photograph. Give each row a breath.

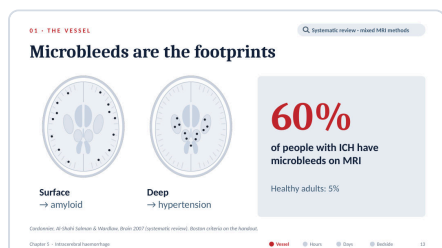
>> "MRI shows us the footprints of these diseases."

🔍 PLAIN WORDS

Read where the bleed is on the scan report. It tells you what to assess first.

🔍 IF CHALLENGED

PLIC: three imaging cohorts. Puig 2019 (n = 43): PLIC plus motor score above 6, AUC 0.89, PPV 65%, NPV 92%. Murray 2025 (98 MISTIE III scans): PLIC overlap on routine CT predicted worse motor outcome, but inter-rater κ was only 0.45. Gupta 2025 (n = 305) held up in a validation cohort. A predictor, not a sentence.



Slide 13

🕒 0:35

13 Microbleeds are the footprints

KEY POINT

Microbleeds on MRI reveal which disease is at work.

SAY

- Microbleeds are tiny old leaks of blood: **black dots** on MRI.
- **Surface** dots point to amyloid. **Deep** dots point to hypertension.
- Charlotte Cordonnier, **one of this chapter's authors**, found them in **60 percent** of people with ICH, against 5 percent of healthy adults.

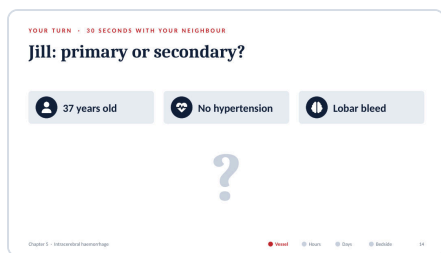
>> "Your turn."

🔍 PLAIN WORDS

Black dots on MRI are old, tiny bleeds.

🔍 IF CHALLENGED

The 60% pools studies with different MRI sequences and field strengths. Newer sequences find more, so the figure depends on the scanner.



Slide 14

🕒 0:45

14 Jill: primary or secondary?

KEY POINT

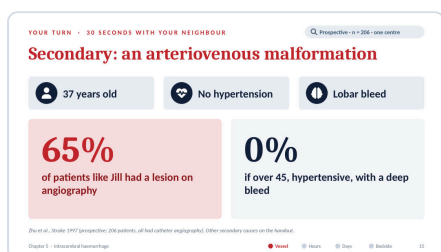
Let the room work out Jill's diagnosis. Soft break 1, at about minute 8.

SAY

- Jill was **37**. She had **no history of high blood pressure**. Her bleed was **lobar**.
- Primary small-vessel disease, or a secondary cause? **Turn to the person next to you. Thirty seconds.**

👉 **ASK THE ROOM** Stop talking. Count to 30 in your head, then ask two people for their answer. Online: "type primary or secondary in the chat".

» Advance: "Secondary."



Slide 15

🕒 0:40

15 Secondary: an AVM

KEY POINT

Jill's profile was a textbook warning sign for a hidden lesion.

SAY

- **Secondary.** Jill had an **arteriovenous malformation**, a tangle of vessels with no capillaries in between.
- In 206 patients who all had angiography, **65 percent** of young, normotensive patients with a lobar bleed had a lesion. Over 45, hypertensive, deep bleed: **zero**.
- **The patient's story tells you where to look.**

» "Stage two."

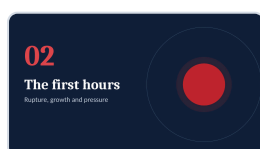
Q IF CHALLENGED

One Hong Kong centre, 1997, catheter angiography in everyone. CT angiography has since changed screening, but the pattern holds.

02 The first hours

Rupture, growth and pressure. The golf-ball moment lives here.

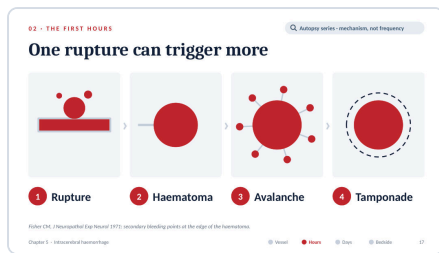
🕒 3:25



Slide 16

🕒 0:05

- "Stage two. **The first hours.** Jill had four of them."



Slide 17

🕒 0:35

17 One rupture can trigger more

KEY POINT

A bleed keeps spreading by tearing the vessels around it.

SAY

- **One:** a diseased vessel ruptures. **Two:** blood splits the tissue and a haematoma forms.
- **Three** is the surprise. Miller Fisher, 1971, found extra bleeding points at the edge: the growing clot tears its neighbours. An **avalanche**.
- **Four:** the bleeding stops when tissue pressure and clotting finally win.

➤ **POINT** Move left to right across the four pictures.

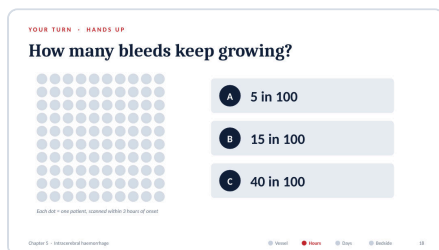
» "So how often does the bleeding keep going?"

PLAIN WORDS

The growing clot tears the small vessels around it, and they bleed too.

IF CHALLENGED

Fisher's evidence is autopsy: it shows a mechanism, not how often it happens. The CT data on the next slide show how often.



Slide 18

🕒 0:30

18 How many bleeds keep growing?

KEY POINT

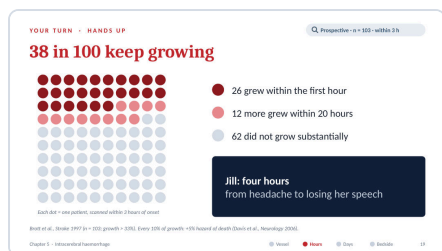
Get the room to commit to a guess. Soft break 2, at about minute 10.

SAY

- Here are **100 patients**, all scanned within three hours of onset.
- How many bleeds keep growing? **A, five. B, fifteen. C, forty.**

👉 **ASK THE ROOM** Hands up for each option. Say what you see: "Most of you went for B."

» *Advance: the dots colour in.*



Slide 19

🕒 0:50

19 38 in 100 keep growing

KEY POINT

38 in 100 bleeds keep growing, and growth predicts death.

SAY

- **Thirty-eight. 26** grew by more than a third within an hour of the first scan, **12 more** within twenty hours, with neurological decline.
- This is **Jill's four hours**. Every 10 percent of growth raised the risk of death by **5 percent**.
- For us, this is why **neuro observations come before any handling** in the first day.

🔊 **PAUSE** Wait for the dots to finish colouring before you speak.

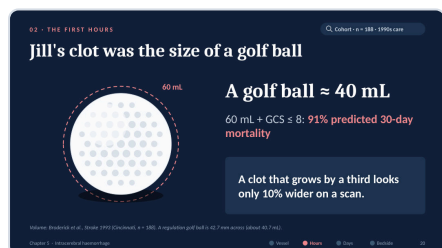
» "Jill's surgeons found out how big hers got."

🗣️ PLAIN WORDS

More than a third of bleeds keep growing in the first day.

🔍 IF CHALLENGED

Brott: n = 103 within 3 h; growth defined as more than a third. Later studies use 6 mL or 33%, so rates vary with the definition and scan timing.



Slide 20

🕒 0:50

20 Jill's clot was the size of a golf ball

KEY POINT

Small changes in size are deadly, and easy to miss. Your jaw-dropping moment.

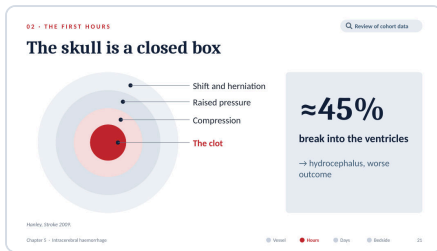
SAY

- Jill's surgeons removed a clot **this size**. A golf ball: about **40 millilitres**.
- Grow it to 60, the dashed line. A clot that grows by a third looks only **10 percent wider** on a scan. The patient feels every millilitre.
- At 60 millilitres with a GCS of 8 or less, predicted 30-day mortality was **91 percent**.
- **PROP** Take the golf ball out BEFORE you speak. Hold it at shoulder height; three seconds of silence. Put it down where everyone can still see it.

» "Why does volume matter so much?"

🔍 IF CHALLENGED

1990s Cincinnati care, when early withdrawal of support was common: scores can become self-fulfilling prophecies (Becker 2001). Use them to inform, never to decide.



Slide 21

🕒 0:35

21 The skull is a closed box

KEY POINT

Inside a closed skull, extra volume becomes extra pressure.

SAY

- **The skull is a closed box.** As the clot grows, pressure rises; if it keeps rising, the brain shifts and can **herniate**.
- About **45 percent** of bleeds break into the ventricles and can block CSF: **hydrocephalus**.
- At the bedside, hydrocephalus looks like **drowsiness, headache and vomiting**. Hold that thought.

🚩 **POINT** Trace the rings from the centre outwards.

>> "Stage three."

🗣️ PLAIN WORDS

Hydrocephalus at the bedside:
drowsy, headache, vomiting.

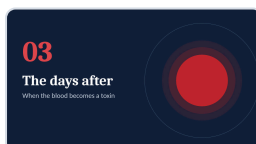
🔍 IF CHALLENGED

The ~45% comes from a review of cohort data. Rates vary with how early and how often patients are scanned.

03 The days after

How the blood itself keeps injuring the brain.

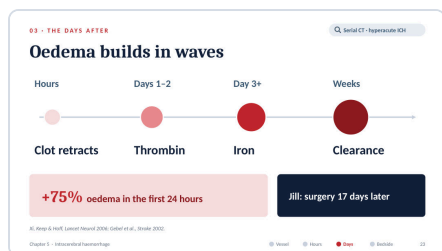
🕒 2:50



Slide 22

🕒 0:05

- "Stage three. **The bleeding has stopped. The injury has not.**"



Slide 23

⌚ 0:50

23 Oedema builds in waves

KEY POINT

Swelling builds in waves for up to two weeks.

SAY

- **Hours:** the clot retracts and squeezes serum out. **Days 1–2:** thrombin opens the blood–brain barrier.
- **Day 3 on:** red cells break down and release iron. **Weeks:** scavenger cells clear the clot.
- Oedema grew **75 percent** in the first 24 hours. **The days after are not a safe plateau.** Jill's surgery came 17 days later.

➤ **POINT** Move along the timeline, left to right.

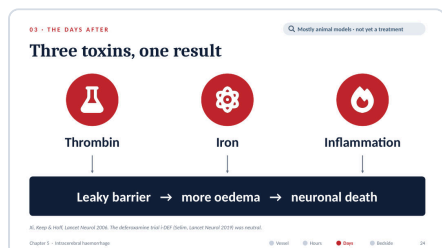
» "Three toxins drive this second wave."

🔍 PLAIN WORDS

Swelling keeps building for days. Stable on day 1 does not mean stable on day 4.

🔍 IF CHALLENGED

The 75% comes from serial CT in hyperacute patients. Measuring oedema on CT is imprecise, and oedema can keep growing into the second week.



Slide 24

⌚ 0:45

24 Three toxins, one result

KEY POINT

Thrombin, iron and inflammation keep injuring the brain after the bleeding stops.

SAY

- **Thrombin** first: needed to stop the bleed, toxic at high levels. **Iron** later, from burst red cells. **Inflammation** throughout.
- All three end in the same place: leaky barrier, more swelling, dead neurons. **That is why patients get worse days later.**

➤ **POINT** Point to the dark bar at the end.

» "True or false?"

🔍 PLAIN WORDS

The blood itself is toxic to the brain.

🔍 IF CHALLENGED

Mostly rat and pig data. The best human test so far, i-DEF with deferoxamine (Selim 2019), was neutral. Mechanism is not yet treatment.

YOUR TURN - TRUE OR FALSE?

True or false?

"The rim around the clot is starving tissue."

Chapter 5 - Intracerebral haemorrhage

Visual | Heart | **Brain** | Bedside

Slide 25

🕒 0:30

25 True or false?

KEY POINT

Test a belief many clinicians hold. Soft break 3, at about minute 14.

SAY

- "The dark rim around the clot is **starving tissue**, like the penumbra in an ischaemic stroke." **Hands up for true.**
- For years people worried that lowering blood pressure would starve that rim.

👉 **ASK THE ROOM** Count the hands and say what you see.

» Advance: "False."

YOUR TURN - TRUE OR FALSE?

False: the rim is resting, not starving

"The rim around the clot is starving tissue."

Lowering BP below 150 did not reduce flow around the clot (ICH ADAPT).

Rim vs the same area on the other side (PET)

Measurement	Other side	Rim around the clot
Blood flow	100%	56%
Oxygen use	100%	48%

Source: Wolf et al., Cereb Blood Flow Metab 2003; 23: 1-10 (after Wolf); Buchner et al., Stroke 2013.

Chapter 5 - Intracerebral haemorrhage

Visual | Heart | **Brain** | Bedside

Slide 26

🕒 0:40

26 False: the rim is resting, not starving

KEY POINT

The rim around the clot is resting, not starving.

SAY

- **False.** PET in 19 patients: blood flow in the rim **56 percent**, oxygen use even lower, **48 percent. Resting, not starving.**
- In ICH ADAPT, lowering systolic pressure **below 150** did not reduce flow around the clot.
- For us: follow the team's **blood pressure targets** during activity. Pressure still matters for growth.

👉 **POINT** Point to each pair of bars in turn.

» "So what does this mean for physiotherapy?"

Q IF CHALLENGED

PET n = 19 shows a mechanism. ICH ADAPT (n = 75) tested it, but its endpoint was CT perfusion, not function.

04 At the bedside

When to start, a Day-4 case, the recovery debate, then close on Jill.

🕒 5:50

04

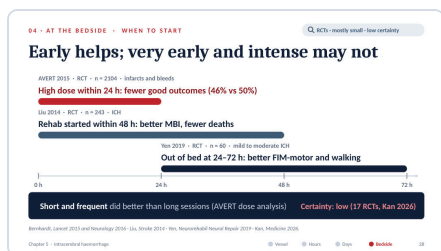
At the bedside

What it means for physiotherapy

Slide 27

🕒 0:10

- "What does this mean for **physiotherapy**? Let's follow the patient through time: the first hours, the days after, the months of recovery."



Slide 28

🕒 1:00

28 Early helps; very early and intense may not

KEY POINT

Start early with short, frequent sessions, but not big in the first 24 hours.

SAY

- **AVERT**, over 2000 patients, infarcts and bleeds: high-dose mobilisation within 24 hours gave **fewer good outcomes**, 46 against 50 percent.
- **Liu**, 243 patients with ICH: rehab within 48 hours, better function at six months and fewer deaths. **Yen**, 60 patients: out of bed at 24 to 72 hours, better motor function and walking, with no blood pressure instability.
- Within AVERT, **shorter, more frequent sessions** did better. And 17 trials pooled in 2026: **certainty low**.
- So: **early, short and frequent**, once the team says the patient is stable.

👉 **POINT** Point to each bar in turn: red, then slate, then navy, then the dark strip.

» "Now let's put it together."

PLAIN WORDS

Start early, keep sessions short and frequent, and don't go big in the first 24 hours.

IF CHALLENGED

Liu HR 4.44, 95% CI 1.24-15.87: few deaths, so a signal, not an effect size. AVERT's ICH subgroup leaned towards usual care with no significant interaction. The dose analysis is observational within the trial (Class III). In Kan 2026, safety outcomes were too poorly reported to pool.

YOUR TURN - CLINICAL REASONING

Day 4: drowsier than yesterday

MR K, 68 - A COMPOSITE CASE

Left putaminal bleed, 22 mL

Long-standing hypertension

Yesterday: sat out 20 min, GCS 15

TODAY, 9 A.M.

- GCS 13
- New headache
- Vomited once
- BP 184/98

Planned: sit-to-stand practice. What do you do?

A Go ahead, but shorter B Bed-level work only C Hold and escalate

Chapter 5 - Intracerebral haemorrhage

Slide 29

🕒 1:00

29 Day 4: drowsier than yesterday

KEY POINT

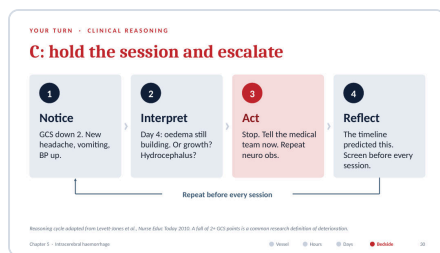
A bedside decision that the pathophysiology predicts. Soft break 4, at about minute 16½.

SAY

- Mr K, 68, hypertensive: a **left putaminal bleed, 22 millilitres**, about half a golf ball. Yesterday he sat out for 20 minutes, GCS 15.
- Today at 9 a.m.: **GCS 13, new headache, vomited once, BP 184 over 98**. You'd planned sit-to-stand practice.
- **A**: go ahead, but shorter. **B**: bed-level work only. **C**: hold and escalate.

👉 **ASK THE ROOM** Hands up for A, B or C (or the chat online). Say what you see.

» Advance: "C."



Slide 30

🕒 0:50

30 C: hold the session and escalate

KEY POINT

New drowsiness after a bleed is the bleed talking until proven otherwise.

SAY

- **Notice:** GCS down two, plus headache, vomiting and higher BP. A two-point fall is a common research definition of deterioration.
- **Interpret:** day 4, oedema still building. Or growth, or hydrocephalus, the closed box. Working out which is the medical team's job.
- **Act:** stop, tell the team now, repeat the obs. **Reflect:** the timeline predicted this. **"Not today" is a clinical decision, not a failure.**

🚩 **POINT** Walk the four steps, then trace the return arrow.

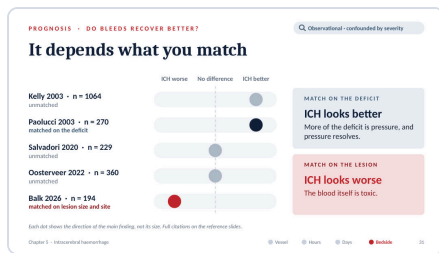
» *"Families ask us about recovery."*

🗣️ PLAIN WORDS

New drowsiness, headache or vomiting: don't start. Tell the team now.

🔍 IF CHALLENGED

A fall of 2+ GCS points is one research definition of early neurological deterioration; others use a rise of 4+ on the NIHSS. The cycle is adapted from Levett-Jones et al. 2010.



Slide 31

🕒 1:10

31 It depends what you match

KEY POINT

Whether bleeds recover better depends on how the comparison is matched.

SAY

- Do bleeds recover better than infarcts? **Unmatched studies disagree:** Kelly found bigger gains; two newer studies, no difference.
- **Paolucci matched on the deficit:** bleeds did better, 2.5 times the odds of a big gain. **Balk matched on lesion size and site:** bleeds did worse.
- Both fit Chapter 5. More of a bleed's deficit is **pressure, which resolves**. But the **blood itself is toxic**.
- So: don't promise a better recovery, and **don't set a ceiling early**. After severe deep bleeds, most motor gains came in the first four months.

🚩 **POINT** Grey dots first, then the dark dot, then the red dot.

🐢 **SLOW DOWN** Land "It depends what you match" slowly.

» "So every stage is a window."

PLAIN WORDS

Don't promise a better recovery because it's a bleed. Don't give up early either.

IF CHALLENGED

All observational, survivors only (about 40% die). ICH patients start lower, so they have more room to gain on Barthel and FIM, which have ceiling effects. Paolucci OR 2.48 (1.19–5.20); Balk mRS OR 1.69 (1.02–2.79) favouring infarct; Yoo 2026 review of 25 studies.



Slide 32

🕒 0:45

32 Every stage is a window to act

KEY POINT

Each stage of the process gives physiotherapy a job. The heart of the talk.

SAY

- **The vessel:** know the cause. Check BP before, during and after activity.
- **The first hours:** neuro obs first. Short, gentle sessions; nothing intense.
- **The days after:** screen before every session; expect ups and downs; **drowsy means escalate**.
- **The months after:** no early ceiling. Target what the location predicts.

🐢 **SLOW DOWN** Slow right down. Give each row its own breath.

» "If you remember three things from tonight..."

SUMMARY

If you remember three things

1

The vessel decides where it bleeds.

2

The first hours decide how big it gets.

3

The blood keeps injuring the brain for days.

Chapter 5 · Introduction to Neuroscience

Vessel
Hours
Days
Stroke
22

Slide 33

0:20

33 If you remember three things

WORD FOR WORD

KEY POINT

The three messages, one last time.

SAY

- The **vessel** decides where it bleeds.
- The **first hours** decide how big it gets.
- The **blood** keeps injuring the brain for days.

PAUSE Pause between each line.

>> "Back to Jill."

EIGHT YEARS LATER

Jill walked onto the TED stage holding a human brain.

ICH is a process, not a moment.

Which claim would you challenge?

Slide 34

0:35

34 Eight years later

WORD FOR WORD

KEY POINT

Close the story, land the big idea and invite challenge.

SAY

- Back to Jill. It took her **eight years** to recover. Then she walked onto the TED stage holding a real human brain.
- Her bleed was not a moment. It was a process, and so was her recovery.
- ICH is a process, not a moment.**
- Thank you. **Which claim tonight would you challenge?**

PAUSE Three full seconds after the big idea. If the room is quiet, wait five seconds, then ask the back-up question below.

>> Questions.



Questions

Leave the references up while you take questions.

SOURCES

References: the chapter story

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15. Broderick JP, et al. JAMA 2001;286:2703-2706.

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35 References: the chapter story

KEY POINT

Leave this up while you take questions.

SAY

- Every figure comes from these papers. **The first reference is the chapter itself.**

SOURCES

References: physiotherapy evidence

1. Borchardt L, et al. *ANESTHESIA* 2015; doi:10.1016/j.anesth.2015.04.005
2. Borchardt L, et al. *ANESTHESIA* 2015; doi:10.1016/j.anesth.2015.04.005
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36 References: physiotherapy evidence

KEY POINT

The physiotherapy trials and cohorts, with DOIs.

SAY

- Go here if someone asks for a source.
- Hidden backups: type **37** and press Enter for other secondary causes, or **38** for what has changed since 2014.

Slide 36



Likely questions

Answer in under 30 seconds: answer, one fact, stop.

Liu's hazard ratio of 4.44 looks huge. Do you believe it?

As a signal, not an effect size. The 95% CI runs from 1.24 to 15.87, with few deaths, one country, and usual care that may have started rehab late.

AVERT was mostly infarcts. Why apply it to ICH?

It included bleeds, and the ICH subgroup leaned towards usual care, though the interaction was not significant. So: caution in the first 24 h, not a ban. Yen started at 24–72 h, outside AVERT's window.

What blood pressure should stop a session?

Use the medical team's targets for that patient. No ICH trial gives a physiotherapy threshold. Yen 2021 found early mobilisation did not raise BP variability.

Is PLIC involvement on CT reliable?

Moderately. Inter-rater κ was 0.45 in Murray 2025, better with expert readers. Combine it with the motor exam: in Puig 2019, PLIC plus motor score gave AUC 0.89.

Do bleeds recover better than infarcts?

It depends on matching. Matched on deficit (Paolucci): better. Matched on lesion size and site (Balk): worse. Both fit the pathophysiology.

Why trust animal data on thrombin and iron?

We shouldn't fully. It explains the timing of decline, but i-DEF (deferroxamine) was neutral. Mechanism is not yet treatment.

Is the 40% figure out of date?

It covers studies to 2008. Later data show modest gains, and early withdrawal of care can inflate it (Becker 2001).

Why do patients worsen on days 3–5?

Oedema still building, iron release, hydrocephalus; also pneumonia or seizures. Screen before every session.

Which outcome measures would you use?

Match the measure to the question: FAC for walking, Fugl-Meyer for motor impairment, PASS for early postural control. Barthel and FIM have ceiling effects.

How do you confirm amyloid angiopathy?

In life, the Boston criteria on MRI (lobar microbleeds, superficial siderosis); v2.0 (2022) added white-matter markers.

Is lowering BP risky for the rim?

No. The rim is resting, not starving (PET). ICH ADAPT found below 150 did not reduce flow; INTERACT2 found below 140 safe.

A question you can't answer

"Good question. I don't want to guess. I'll check the source and come back to you."

Q When someone challenges a claim

1

Thank them. "Good challenge." It buys you two seconds.

2

Name the design and its limit. "That's one RCT of 243, with a wide interval."

3

Give your position. "So I read it as a signal, not an effect size."

4

Stop. Or: "Can I check that and come back to you?"

🔊 If the room goes quiet

- Ask: "Which of these windows do we meet most often in our own practice, and are we timing our assessments around it?"
- Then wait a full five seconds. Silence gives people time to think.